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1.2.1: What is a state-space model and why do I need to know about them?

1.2.2: Example continuous-time state-space models used for tracking applications

✓ Reading: Notes for Lesson 1.2.2

10 min

📺 Video: 1.2.2: Example continuous-time state-space models used for tracking applications

24 min

✓ Reading: Introducing a new element to the course!

10 min

📄 Lab: Jupyter notebook used in conjunction with practice quiz

15 min

📋 Practice Assignment: Practice assignment for Lesson 1.2.2

10 min

1.2.3: Understanding the time-domain response of a state-space model

1.2.4: Illustrating the time-domain response

1.2.5: Converting continuous-time state-space models to discrete-time

1.2.6: How do I simulate a discrete-time state-space model?

1.2.7: Is it even possible for a Kalman filter to estimate this model's state?

1.2.8: Summary of "What do I need to know about state-space models?" module plus next steps

Introducing a new element to the course!

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Until now, the development of state-space models has been entirely mathematical in nature. From now on, we begin to look at how to use this mathematical model in computer programs. We begin with simulations of a few commonly encountered state-space models to help build your intuition regarding how they work.

In this course, you will use Octave to implement algorithms. If you are already familiar with Octave, great! If you are already familiar with MATLAB, then you will have no difficulty with Octave since its basic syntax is the same. If you have never used Octave or MATLAB, then I recommend that you seek out tutorials to aid your learning. The MathWorks has some excellent web-based tutorials (including videos) for MATLAB (<https://www.mathworks.com/support/learn-with-matlab-tutorials.html> [↗](#)), and since Octave's syntax is almost the same, these tutorials will be very helpful. You can also find an Octave tutorial at https://en.wikibooks.org/wiki/Octave_Programming_Tutorial. [↗](#)

You do not need to download Octave to your computer. All of the programming assignments can be completed inside your web browser using an interactive web-based "Coursera Labs Jupyter notebook." This notebook tool will allow you to enter code, execute code, and see results. Sometimes, you will use Octave code in a notebook to answer quiz questions. When this is the case, I recommend that you have two tabs (or windows) open, with the Octave code in one tab and the notebook in the other tab, to allow easy switching between the computational tool and the form for entering quiz answers.

In this lesson, you have already watched a video that describes how some Octave code designed to implement different state-space models works. Next, you will execute an Octave notebook to try some of the commands implemented in this Octave code yourself. The practice quiz will test your mastery of this particular sample notebook by asking you to enter results that it computes as quiz answers.

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